Minh Hai NGUYEN

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I am a 3rd-year PhD candidate in MAMBO team at Toulouse university (expected graduation: Dec. 2026), advised by Prof. Pierre Weiss and Prof. Edouard Pauwels. My research focuses on computational imaging with deep learning and generative models. I also actively contribute to open-source software.

EDUCATION

PhD in Applied Mathematics — 2023 -
Toulouse University, France
Advised by Prof. Pierre Weiss and Prof. Edouard Pauwels
Research topics: Inverse Problems in Imaging, applications to Microscopy

Master of Engineering in Applied Mathematics — 2018 - 2023
INSA Toulouse, France
Major in AI / Data Science
(*Location: 2018-2020 in Vietnam and 2020-2023 in France)

EXPERIENCE

Research Engineer — Dec. 2022 - Sept. 2023
Agenium Space
Correcting micro-vibrations in satellite imaging with pushbroom cameras.

Research Intern — Jun. 2022 - Sept. 2022
Institut Mathématiques de Toulouse
Product-convolution neural networks: applications to image deblurring with space-varying blurs.

PUBLICATIONS

[1] J. Tachella et al. *DeepInverse: A Python package for solving imaging inverse problems with deep learning*. 2025. arXiv: `2505.20160 [eess.IV]`.

[2] M. H. Nguyen, E. Pauwels, and P. Weiss. “How Diffusion Prior Landscapes Shape the Posterior in Blind Deconvolution”. In: *Pre-print* (2024).

[3] M. H. Nguyen, F. de Vieilleville, and P. Weiss. “DeepVibes: Correcting Microvibrations in Satellite Imaging With Pushbroom Cameras”. In: *IEEE Transactions on Geoscience and Remote Sensing* 62 (2024), pp. 1–9.

[4] M. H. Nguyen and P. Weiss. “Comparing Plug-and-Play and Unrolled networks”. In: *Pre-print* (2024).

LANGUAGE

English: fluent **French:** fluent **Vietnamese:** native

PROGRAMMING LANGUAGE

Python (advanced): PyTorch (DDP, AMP), Jax, scientific computing **C/C++ (intermediate)**

OTHER SERVICES

Open source project

DeepInverse — Github Repository
PyTorch library for solving imaging inverse problems using deep learning

Teaching

- Signal Processing (1st Master level) at INSA Toulouse, in 2024.
- Stochastic Optimization and Learning (1st Master level) at INSA Toulouse / ENSEEIHT, in 2024.

HONOR AND AWARDS

French National Prize for Open Science Library for DeepInverse. Nov. 2024
Documentation prize
By Ministère de l'Enseignement Supérieur et de la Recherche

Doctoral Research Fellowship, French Ministry of Higher Education and Research 2023

Fully Funded Doctoral Research Grant (2023-2026)

IA Pau Competition 2023
Best project award
By IA Pau organization, funded by Boavizta (1500eur prize).